

SentinelOps Weekly Progress Report

Sample report only. This file previews the weekly report format before Sprint 1 begins and should not be treated as a submitted sprint progress report.

Report Information

Field	Value
Course	SWENG 894 - Software Engineering Capstone Experience
Project	SentinelOps
Sprint	Pre-Sprint / Planning
Reporting Week	Week 0 Sample
Reporting Window	2026-05-11 to 2026-05-17
Student	Eli Vazquez
Report Date	2026-05-17
Repository	SentinelOps GitHub repository
Jira Board	https://psu-capstone-sentinelops.atlassian.net/jira/software/projects/SCRUM/boards/1/backlog

Sprint Goal

Sprint 1 has not started yet. The intended Sprint 1 goal is to establish the project foundation for the first predictive maintenance workflow: telemetry generation, raw data storage, initial feature processing, and Airflow-based orchestration.

The practical objective is to produce a small but traceable vertical slice that can be expanded in later sprints without forcing a redesign of the API, Spark processing, or orchestration boundaries.

Sprint Planning

Sprint Backlog

ID	Requirement / User Story	Priority	Estimate	Status
FR-01	Telemetry Generation and Ingestion	High	5 SP	Planned
FR-02	Raw Telemetry Storage	High	3 SP	Planned
FR-03	Feature Engineering Processing	High	5 SP	Planned
FR-04	Workflow Orchestration	High	8 SP	Planned

ID	Requirement / User Story	Priority	Estimate	Status
NFR-01	Failed Workflow Detection and Reporting	High	Not estimated	Planned
NFR-03	Component Responsibility Separation	High	Not estimated	Planned
NFR-04	Repeatable Local Execution	Medium	Not estimated	Planned

Product Backlog Updates

The functional and non-functional requirements have been documented and loaded into Jira. Sprint assignments have been created for four planned sprints, with Sprint 1 focused on telemetry, storage, feature processing, and orchestration foundations.

No implementation-driven backlog changes have occurred yet because the sprint has not started.

Definition of Done

Area	Definition of Done Criteria
Requirements	Acceptance criteria reviewed and updated when implementation clarifies behavior
Design	Architecture notes updated when component boundaries change
Development	Source code implemented, committed, and traceable to a requirement
Testing	Practical validation added for implemented behavior
Documentation	Relevant setup, architecture, or usage documentation updated
Integration	Changes run locally through the intended workflow where feasible
Review	Work checked against acceptance criteria before marking complete

Acceptance Criteria Review

FR-01 - Telemetry Generation and Ingestion

Given	When	Then
The telemetry simulator is configured	A telemetry generation workflow is executed	Representative telemetry data is produced and available for processing

FR-02 - Raw Telemetry Storage

Given	When	Then
Telemetry data is generated or ingested	The ingestion workflow completes	Raw telemetry data is persisted in the configured storage location

FR-03 - Feature Engineering Processing

Given	When	Then
Raw telemetry data exists	The feature engineering workflow executes	Processed feature sets are generated and stored for predictive scoring

FR-04 - Workflow Orchestration

Given	When	Then
Workflow definitions are configured	A workflow is triggered	Telemetry ingestion, processing, scoring, and reporting tasks execute in the defined sequence

Source Code Development

Summary of Contributions

This sample report reflects planning work rather than sprint implementation. Current progress includes documentation structure cleanup, requirements documentation, architecture documentation, Jira backlog creation, and sprint assignment.

No Sprint 1 source-code work is reported here because the sprint has not started.

Repository Information

Resource	Link
Git Repository	SentinelOps GitHub repository
Jira Board	https://psu-capstone-sentinelops.atlassian.net/jira/software/projects/SCRUM/boards/1/backlog
Sprint Board	Jira board Sprint 1 view

Important Commits

Commit ID	Description	Related Requirement
cca5e69	Merge project conception and documentation updates	Documentation
5a815f0	Complete project conception documentation	Documentation
fa618ec	Initialize project structure and CONOPS documentation	Documentation

Burndown Status

Sprint 1 has not started, so no burndown trend is available yet. The initial sprint backlog contains 21 functional story points plus supporting non-functional requirements related to reliability, maintainability, and local execution.

Once Sprint 1 begins, this section should summarize completed points, remaining points, scope changes, and any delivery risk observed during the reporting window.

Burndown Chart

No burndown chart is included in this sample because Sprint 1 has not started.

Software Testing

Testing Overview

No implementation tests are reported for this pre-sprint sample. Early Sprint 1 testing should focus on validating telemetry generation, persistence, feature processing, and workflow execution behavior.

Requirement-to-Test Mapping

Requirement ID	Test Case ID	Test Description	Status
FR-01	TC-01	Validate telemetry generation produces representative asset telemetry	Planned
FR-02	TC-02	Validate raw telemetry is persisted after ingestion	Planned
FR-03	TC-03	Validate feature processing creates structured feature output	Planned
FR-04	TC-04	Validate Airflow workflow executes tasks in the expected sequence	Planned
NFR-04	TC-05	Validate the local setup can be repeated from documented instructions	Planned

Test Case Specifications

TC-01 - Validate Telemetry Generation

Field	Description
Related Requirement	FR-01
Preconditions	Telemetry simulator configuration exists
Test Steps	Execute the simulator or telemetry generation workflow
Expected Result	Representative telemetry records are generated for sample assets
Status	Planned

TC-04 - Validate Workflow Orchestration

Field	Description
Related Requirement	FR-04

Field	Description
Preconditions	Airflow DAG and local services are configured
Test Steps	Trigger the predictive maintenance workflow
Expected Result	Workflow tasks execute in the documented sequence and expose execution status
Status	Planned

Risks and Roadblocks

- Airflow and Spark integration should be kept intentionally small in Sprint 1 so the first workflow remains demonstrable.
- The data model for telemetry and processed features may need adjustment after the first implementation pass.
- Test scope should stay tied to implemented behavior to avoid creating brittle placeholder tests.

Planned Activities for Next Week

- Confirm Sprint 1 backlog readiness in Jira.
- Start the telemetry simulator implementation.
- Define the first raw telemetry storage format.
- Create the initial feature-processing path.
- Draft the first Airflow DAG for the telemetry processing workflow.
- Add practical validation for the first implemented workflow steps.

Overall Sprint Assessment

The project is ready to enter Sprint 1 planning with a clear backlog and traceable requirements. The highest near-term risk is integration complexity, so the first sprint should prioritize a small end-to-end workflow over broad feature coverage.